

1. The company wants to automate its payroll process.
Write a C program to calculate the total salary by adding the basic salary and bonus amount of employee.

⇒ Input:

Basic Salary

Bonus salary.

Output:

Total Salary

Pseudo code:-

Start

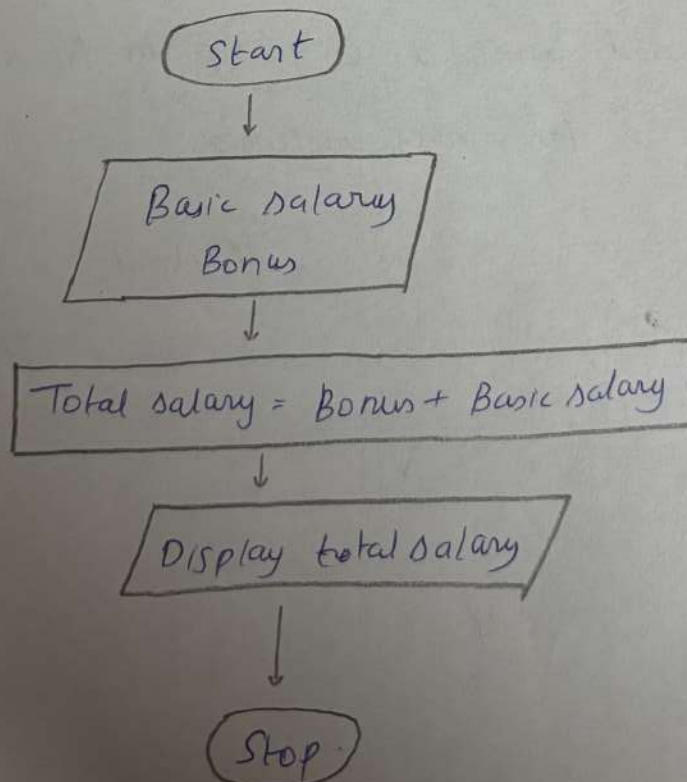
Read Basic salary, Bonus

Total salary = Basic salary + Bonus

Display to the salary.

Stop.

Flow chart:-



C program:-

```
#include <stdio.h>
int main ()
{
    float basic, bonus, total;
    scanf ("%f %f", &basic, &bonus);
    printf ("Enter the salary");
    printf (" Enter bonus amount");
    total = basic + bonus;
    printf (" Total salary = %.2f ", total);
    return 0;
}
```

2. During a database update, two employee IDs need to be exchanged. Write a C program to swap two numbers using a temporary variable.

⇒

Input:

First number (A)

Second number (B)

Output:

Swapped values of A and B.

Pseudo code:-

Start

Read A, B

Temp = A

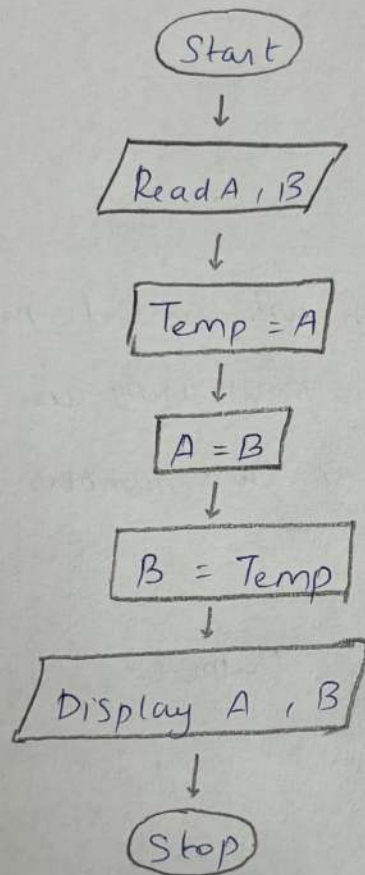
A = B

B = Temp

Display A, B

Stop

Flowchart:-



C program:-

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int A, B, temp;
```

```
printf("Enter two numbers: ");
```

```

scanf ("%d %d", &A, &B);
temp = A; A = B;
B = temp;

printf ("After swapping : \n");
printf ("A = %d \n", A);
printf ("B = %d \n", B);
return 0;
}

```

3. In an embedded system with limited memory, two values must be exchanged without using any extra storage. Write a C program to swap two numbers without using temporary variable.

⇒ Input:

First number (A)

Second number (B)

Output:-

Swapping values of A and B.

Pseudo code:

Start

Read A, B

$A = A + B$

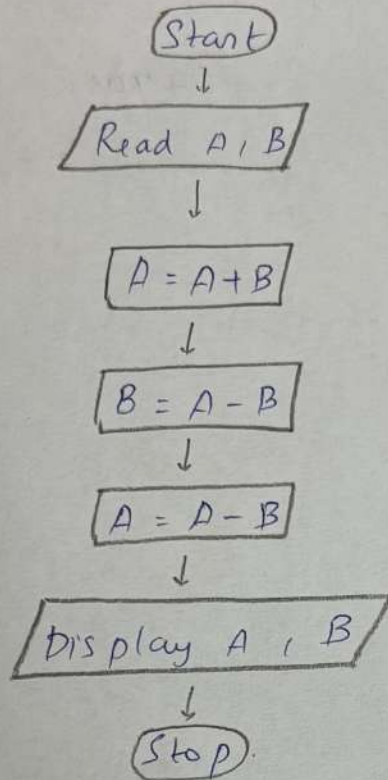
$B = A - B$

$A = A - B$

Display A, B

Stop

Flowchart:-



C program:-

```
#include <stdio.h>
```

```
int main() {
```

```
int A, B; scanf("%d %d", &A, &B);
```

```
A = A + B;
```

```
A = A - B;
```

```
A = A - B;
```

```
printf("After swapping: A = %d B = %d", A, B);
```

```
return 0;
```

```
}
```

4. A bank wants to provide customers with an estimate of the interest earned on their deposits. Write a C program to calculate simple interest using the formula.

⇒

Input:

Principle Amount (P)

Rate of interest (R)

Time (T)

Output:-

Simple interest (SI).

Pseudo code:-

Start

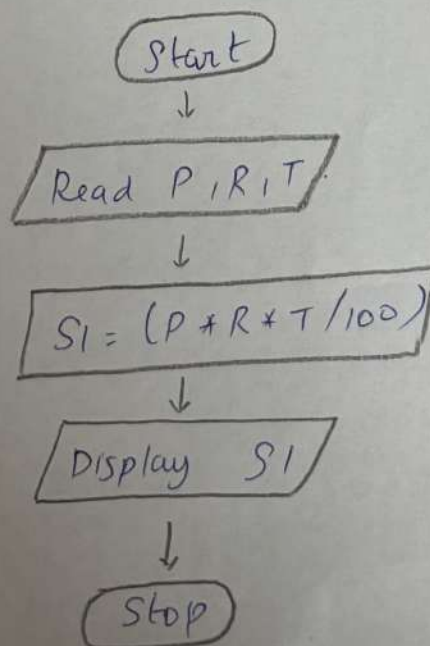
Read P, R, T.

$$SI = (P \times R \times T) / 100$$

Display SI

Stop.

Flow chart:-



C program:-

```
#include <stdio.h>
int main() {
    float P, R, T, SI;
    scanf("%f %f %f", &P, &R, &T);
    printf("Enter the principal amount:");
    printf("Enter the Rate of Interest:");
    printf("Enter the time:");
    SI = (P * R * T) / 100;
    printf("Simple Interest = %.2f", SI);
    return 0;
}
```

5. A weather station records temperature in celcius, but reports need to be generated in Farenheit. Write a C-program to convert temperature from celcius to Farenheit.

⇒ Input:

Temperature in celcius (C)

Output:-

Temperature in Farenheit (F).

Pseudo code:-

Start

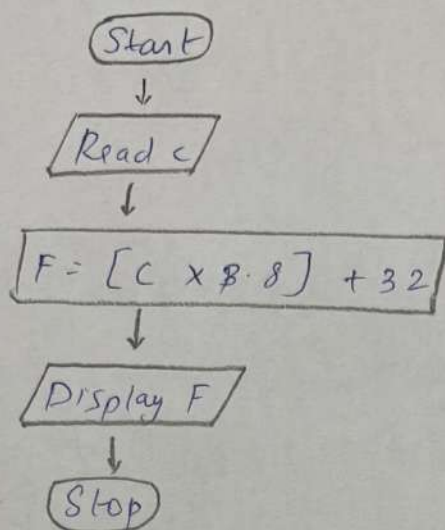
Read C

$F = [C \times 1.8] + 32$

Display F

Stop.

Flow chart .



C program:-

```
#include <stdio.h>
int main(){
    float C, F;
    scanf("%f %f", &C, &F);
    F = [C x 1.8] + 32;
    printf("Temperature in celcius : %.2f", C);
    printf(" Temperature in Fareheit : %.2f", F);
    return 0;
}
```

6. A mathematic learning application needs to display both the quotient and remainder when two numbers are divided. Write a C program to find the quotient and remainder of two integers.

⇒ Input:-

First number (A)

Second number (B)

Output:-

Quotient (Q)

Remainder (R)

Pseudo code:-

Start

Read A, B

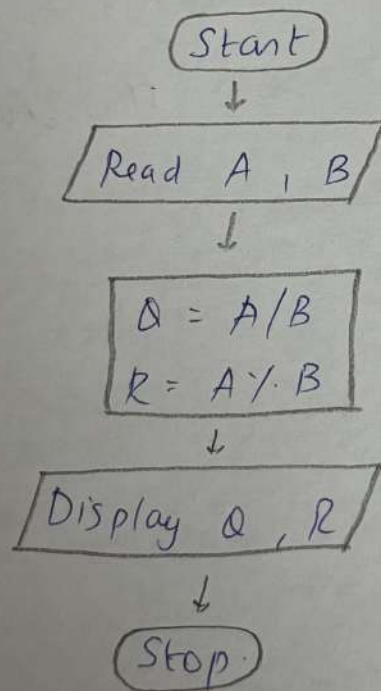
$Q = A / B$

$R = A \% B$

Display Q, R

Stop

Flow chart:-



C program:-

```
#include <stdio.h>
```

```
int main() {
```

```
int a, b;
```

```

scanf ("%d %d", &a, &b);
printf ("Quotient = %d\n", a/b);
printf ("Remainder = %d\n", a%b);
return 0;
}

```

7. An event management system allows entry based on even-numbered ticket IDs. Write a C program to find even or odd.

⇒ Input:-

Ticket Number (N)

Output:-

Even or Odd.

Pseudo code:-

Start

Read N

If $N \% 2 == 0$

Display "Even"

Else

Display "Odd"

End if

Stop

Flow chart:-

